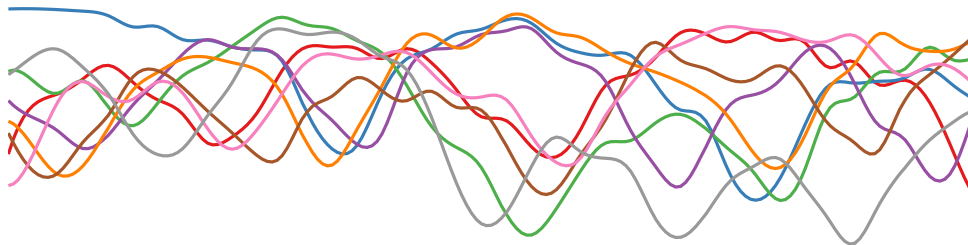


COMPUTATIONAL NEURODYNAMICS

TOPIC 1: INTRODUCTION AND MOTIVATION



Pedro A.M. Mediano

`p.mediano@ic.ac.uk`

(Slides: Pedro Mediano & Murray Shanahan)

INTRODUCTION

STOCKHOLM, OCTOBER 2024



John J. Hopfield



Geoffrey E. Hinton

“For foundational discoveries and inventions that enable machine learning with artificial neural networks.”

— The Nobel Prize in Physics 2024



How can a network of simple neurons *compute*?

COMPUTING WITH NETWORKS OF NEURONS

- ▶ Hopfield showed that a network of many simple, interacting neurons can **store and recall memories**.
- ▶ No single neuron holds a memory. Memories are **attractors** of the collective dynamics of the whole network.
- ▶ The computation is not programmed in: it **emerges** from the interactions between neurons.

Neural networks and physical systems with emergent collective computational abilities.

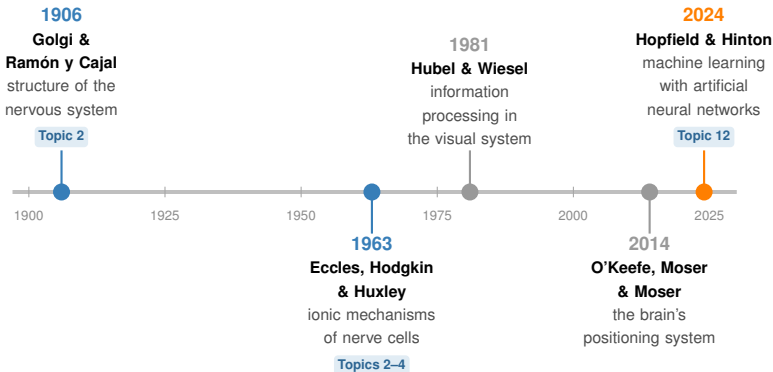
(Title of Hopfield's 1982 paper)

THIS COURSE

How **computation** emerges from the **collective dynamics** of large networks of neurons, and how to simulate and analyse them.

A CENTURY OF NOBEL PRIZES

- Understanding neurons has been rewarded with a Nobel Prize many times...



...and we will meet several of them in this course.

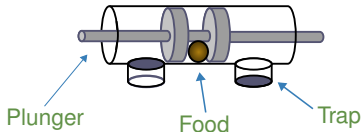
COMPARATIVE COGNITION

COMPARATIVE COGNITION

- ▶ Many animals are capable of impressive cognitive feats despite their lack of language.
- ▶ Cognitive high-fliers include:
 - apes (apart from humans), such as chimpanzees
 - certain birds, especially corvids (rooks, crows, ...)
 - certain cephalopods, notably octopuses
- ▶ Cognitive capabilities include:
 - Tool use and meta-tool use
 - Tool construction
 - Observational learning
 - Metacognition
 - Episodic-like memory

EXAMPLE: PHYSICAL COGNITION

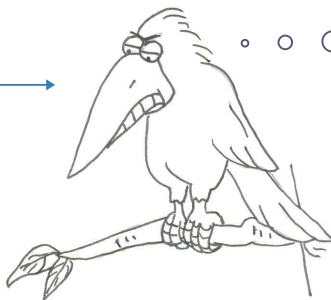
- ▶ It has been shown that rooks and crows are capable of solving problems that require an understanding of the physical properties of objects.
- ▶ One paradigm involves the use of different kinds of [trap-tube](#).



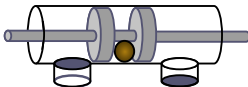
- ▶ The birds must pull the plunger in the right direction to obtain the food, or it is lost in the trap.
- ▶ If a bird spontaneously solves a problem (without trial and error), it is evidence of physical cognition.

THE INITIAL PROBLEM

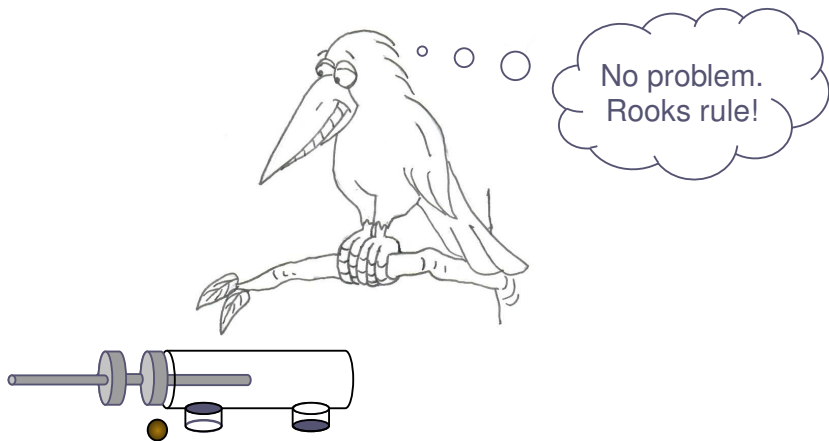
Betty



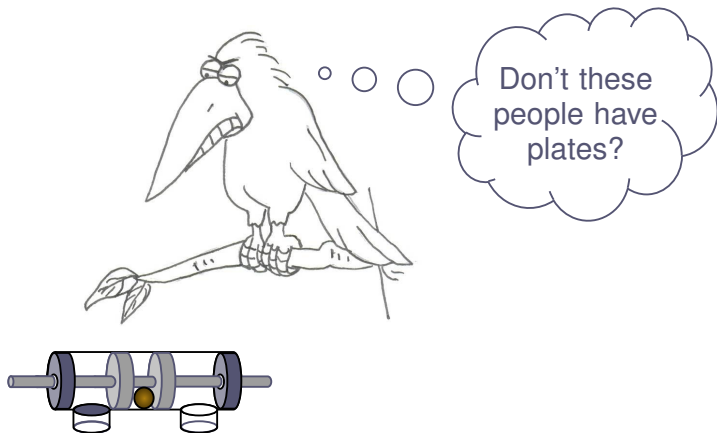
Huh?



SOLVED BY TRIAL AND ERROR



AN UNSEEN VARIATION



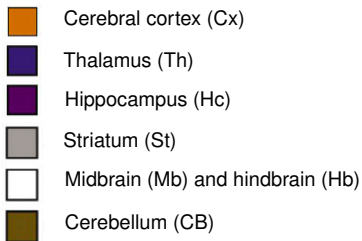
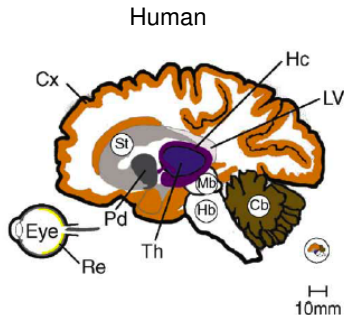
SPONTANEOUSLY SOLVED



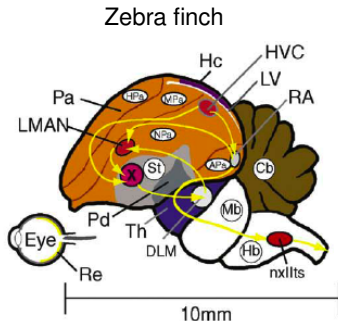
ANIMAL BRAINS

MAMMALIAN BRAINS

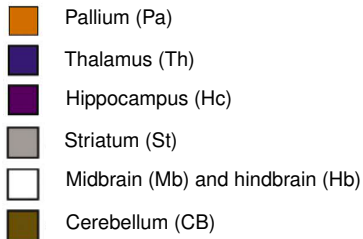
- ▶ The anatomy of the human brain is dominated by its large convoluted cerebral **cortex**.
- ▶ The cerebral cortex is organised into six layers.



AVIAN BRAINS

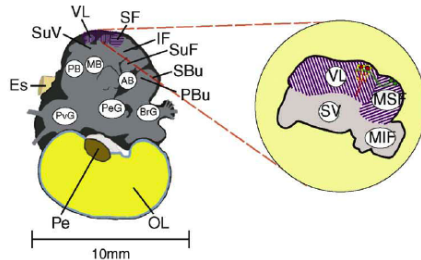


- The homologue of mammalian cortex in a bird's brain is the **pallium**.
- But the pallium is not layered. Rather, it has a nucleated structure.



CEPHALOPOD BRAINS

Octopus



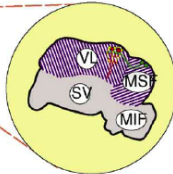
Vertical lobe (VL) and median superior frontal lobe (MSF)



Retina-like optic lobe (OL)

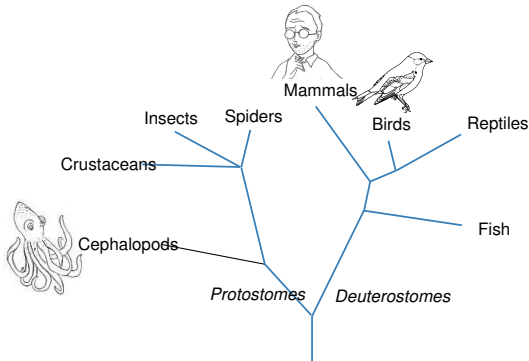


Peduncle (Pe)



- ▶ The octopus brain has almost no structures that are homologous to those of the vertebrate brain.
- ▶ And as well as its central nervous system, it has eight mini-brains (or **ganglia**), one per tentacle.

BRAIN EVOLUTION



- ▶ Octopus and human evolution diverged long before the vertebrate brain appeared.
- ▶ Its brain is made of the same neurons, but its architecture is completely different.

THE SYNTHETIC METHODOLOGY

TOWARDS A DEEP THEORY

- ▶ A deep theory of how cognition (and consciousness) is realised in the biological brain must rest on principles that apply not only to mammals, but also to birds, and to cephalopods.



Here's a good question to ask a neuroscientist: "Does your work help to explain the cognitive capabilities of the octopus?"

- ▶ Even more generally, it should describe the **space of possible minds**:
 - In principle, it should account for cognition as it might evolve elsewhere in the Universe.
 - And it should allow for the construction of forms of artificial cognition.

TOWARDS A DEEP THEORY

- ▶ What form would such a theory take?
- ▶ It must sit at a level above the “lowest common denominator” in the evolution of animal cognition: [the neuron](#).
- ▶ All animal cognition is the product of the dynamics that emerges when large networks of such neurons are connected together.

OUR OBJECT OF STUDY

The [dynamics of large networks of neurons](#), and their role in producing behaviour.

BRAINS ARE COMPLEX SYSTEMS

DEFINITION: COMPLEX SYSTEM

A system comprising large numbers of interacting components, whose individual behaviour is easy to describe mathematically, but whose collective behaviour is not.

- ▶ The human brain has ~ 86 billion neurons. A single neuron can be described by a handful of equations (next topic).
- ▶ But cognition is a property of the **whole network**, not of any single neuron (remember Hopfield's memories).



How do we study a system like this?

TWO WAYS TO STUDY A COMPLEX SYSTEM

THE ANALYTICAL METHOD

Build a mathematical model of the whole network, and **solve** it to predict, or at least characterise formally, its behaviour over time.

- ✓ Exact and general results.
- ✗ Only possible for simple or idealised networks.

THE SYNTHETIC METHOD

Model each component, put many of them together in a computer, and **simulate** the whole system to see what it does.

- ✓ Copes with large, realistic, complex networks.
- ✗ Results are specific, and still need to be understood.



The two complement each other: we will use both.

THE SYNTHETIC METHOD

What I cannot create, I do not understand.

Richard Feynman, 1988

- ▶ Brains are complex systems *par excellence*, so computer simulation is our main tool.
- ▶ We still need mathematical models of the components, i.e. the neurons (Topics 2–4).
- ▶ And we still need mathematical methods to understand the complex behaviour that our simulations produce (Topics 9–11).

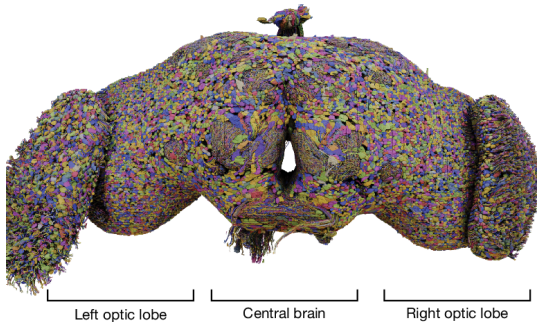


Braitenberg called this **synthetic psychology**: it is easier to build a behaving system than to analyse one (Topic 5).

THE SYNTHETIC METHOD TODAY

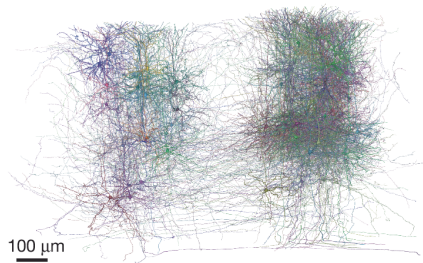
MAPPING EVERY CONNECTION

- To simulate a brain, we first need to know how it is wired. In 2024, the first complete wiring diagram ([connectome](#)) of an adult fruit fly brain was published.



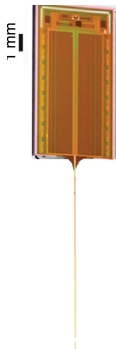
- Every neuron reconstructed from electron microscopy: [139,255 neurons](#), and [54.5 million synapses](#) between them.

MAPPING A CUBIC MILLIMETRE

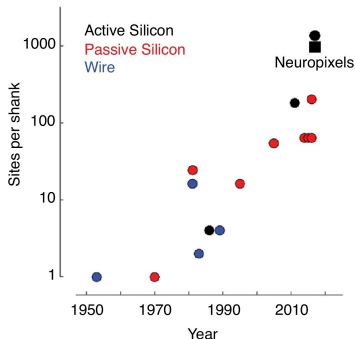


- ▶ MICrONS: about 1 mm³ of mouse visual cortex, with more than 200,000 cells and 0.5 billion synapses.
- ▶ Before slicing it, the activity of ~75,000 of its neurons was recorded in the living animal.
- ▶ A similar map now exists for a cubic millimetre of human cortex: ~57,000 cells and ~150 million synapses (Shapson-Coe et al., 2024).

RECORDING THOUSANDS OF NEURONS



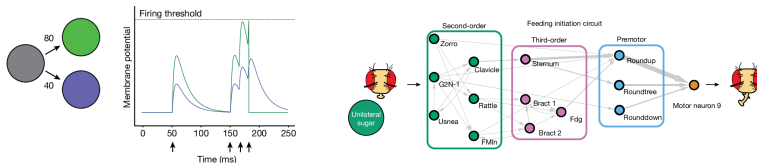
Neuropixels probe (adapted)



- ▶ A single Neuropixels probe has 960 recording sites along a 10 mm shank (Jun et al., 2017).
- ▶ With them, ~30,000 neurons were recorded across 42 regions of the mouse brain (Steinmetz et al., 2019).

SIMULATING A WHOLE BRAIN

- Shiu et al. took the fly connectome, and simulated **every neuron** as a simple leaky integrate-and-fire model.



- Using only the wiring, the model predicted which neurons drive feeding and grooming. **91%** of 164 predictions tested in real flies were correct.

THE SYNTHETIC METHOD, AT SCALE

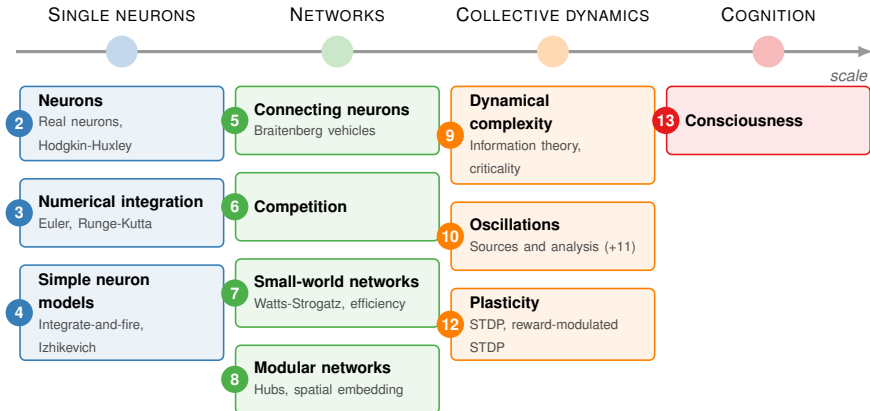
Model the components, wire them as in the real brain, simulate, and compare with the animal.



You will build integrate-and-fire neurons yourself in Topic 4.

COURSE OVERVIEW

COURSE OUTLINE



COURSE OUTLINE

WHO THIS COURSE IS FOR

- ▶ Students interested in interdisciplinary approaches to artificial intelligence.
- ▶ Students interested in the principles behind humans' intelligent behaviour and consciousness.
- ▶ Students considering applications of data analysis in neuroimaging.

WHAT YOU NEED

There are no strict requirements. Some background in programming, statistics, and differential equations is recommended, but not essential.

IMPORTANT INFORMATION

IN-PERSON SESSIONS

Monday 9am, Huxley 311 (lecture).

Friday 9am, Huxley 219 (lab).

TEACHING ASSISTANTS

Hanna Tolle, Max Krahn, Alex Proca

ASSESSMENT

20% Programming coursework (deadline Nov 18th).

80% Final exam.



Materials in Scientia, discussions on EdStem.

PROJECTS AVAILABLE

- ▶ Analysing large-scale brain data via information-invariant measures.
- ▶ Scalable gradient-based inference of brain dynamics for clinical trials.
- ▶ Inferring causal abstractions of computations in recurrent neural networks.
- ▶ Inferring interpretable whole-brain model equations from data.
- ▶ Neural alignment and information flow during psychedelic social interaction.

SUMMARY

- ✓ Many animals with very different brains show impressive cognition.
 - A deep theory must explain the octopus, too.
 - Common denominator: the **dynamics of large networks of neurons**.

- ✓ Brains are complex systems, so we study them **synthetically**:
 - Model the components, simulate the whole, analyse the result.
 - Today, this is possible at the scale of whole (small) brains.

- ✓ **Next up**: neurons, real and artificial.

RELATED READING

Edelman, D.B. & Seth, A.K. (2009). Animal Consciousness: A Synthetic Approach. *Trends in Neurosciences* 32 (9), 476–484.

Seed, A.M. & Byrne, R. (2011). Animal Tool-Use. *Current Biology* 20, R1032–R1039.

Shanahan, M.P. (2010). *Embodiment and the Inner Life: Cognition and Consciousness in the Space of Possible Minds*. Oxford University Press.

Braitenberg, V. (1984). *Vehicles: Experiments in Synthetic Psychology*. MIT Press.

Hopfield, J.J. (1982). Neural networks and physical systems with emergent collective computational abilities. *PNAS* 79(8), 2554–2558.

Shiu, P.K. *et al.* (2024). A *Drosophila* computational brain model reveals sensorimotor processing. *Nature* 634, 210–219.